AAV-Mediated Expression of FLAG-Tagged Tau Protein: A Therapeutic Strategy for Tauopathies

Ishaan Mehrotra,[1*] Shubhika Munot,[1*] Nikki Kulkarni,[1*] Kayla Williams,[1*]Cameron Kim[1†]

1. Department of Biomedical Engineering, Duke University

*These authors contributed equally | Corresponding author: cameron.kim@duke.edu

ABSTRACT

The abnormal aggregation and accumulation of tau protein are hallmark features of several neurodegenerative diseases, collectively known as tauopathies, including Alzheimer's disease (AD), frontotemporal dementia (FTD), and Parkinson's disease (PD). Tau, a microtubule-associated protein, plays a crucial role in maintaining neuronal structure and function. In tauopathies, post-translational modifications (PTMs) alter tau's hydrophilicity, conformation, and stability, leading to neurofibrillary tangles that disrupt neuronal integrity and contribute to disease progression. Current therapies focus on reducing tau levels or preventing its aggregation, providing only modest delays in symptom progression. Thus, a novel approach that mitigates tau toxicity and supports neuronal health is urgently needed. In this study, we explore the potential of adeno-associated virus (AAV)-mediated delivery of functional tau protein to neurons. AAV vectors offer an effective gene delivery platform due to their low immunogenicity, ability to transduce non-dividing cells, and high affinity for neuronal tissue. We successfully cloned the tau transgene into an AAV vector, confirmed through restriction digestion, Sanger sequencing, and whole plasmid sequencing. Quantitative PCR determined a viral titer of 1.5 million viral genomes (vg)/µL, providing standardized dosing for functional assays. Functional analysis via fluorescence imaging demonstrated FLAG-tagged tau expression and aggregation in SKN cells. Tau aggregates, identified by Thioflavin staining, showed partial co-localization with FLAG-tagged tau, particularly at higher multiplicities of infection (MOI 250). Western blot analysis for tau expression was inconclusive due to protein transfer and sample quality issues. Despite this, our results indicate that AAV-based delivery can promote tau expression and aggregation in neuronal cells. This study lays the groundwork for future therapeutic strategies targeting tauopathies, including engineered tau variants resistant to hyperphosphorylation and aggregation, offering a promising approach for neurodegenerative disease treatment.

INTRODUCTION

The abnormal amassing of tau protein is a pathological feature of various tauopathies, such as Alzheimer's disease (AD), frontotemporal dementia (FTD), and Parkinson's disease (PD)[1]. Tau protein, a microtubule-associated protein widely distributed in the central nervous system, plays a critical role in maintaining and regulating neuronal morphology and function[2]. In neurodegenerative diseases, tau protein undergoes post-translational modifications that affect its hydrophilicity, spatial conformation, and stability, promoting abnormal aggregation and the formation of neurofibrillary tangles[3]. These tangles disrupt neuronal structure and function, ultimately leading to neuronal death and contributing to the initiation and progression of neurological disorders. Current therapeutic approaches largely focus on reducing tau levels or preventing further aggregation, which has only served to slightly delay symptom advancement[4]. Hence, there is a need for a novel therapeutic approach that not only mitigates tau-related toxicity but also enhances the overall health and function of affected neurons.

In this study, we aim to utilize adeno-associated virus (AAV) vectors to deliver functional tau protein directly to neurons in an effort to express normal tau function. AAVs are small, non-enveloped viruses that can be genetically engineered to express specific genes of interest. They have emerged as a promising platform for gene delivery to the brain due to their minimal immunogenicity, ability to transduce non-dividing cells, and high affinity to neuronal cells compared to other viral vectors[5]. Our approach builds on previous research that has demonstrated the potential of AAV-mediated gene therapy for neurodegenerative diseases, such as Parkinson’s and Huntington’s, where the delivery of therapeutic proteins has shown promise in preclinical and early clinical studies[5]. However, the application of AAV-based gene therapy, particularly in restoring tau function, remains largely unexplored. While our approach may not immediately restore normal protein activity or reduce aberrant PTMs in unhealthy cells, it represents a critical first step. Successful AAV delivery of functional tau could enable future strategies to deliver mutated tau that are less prone hyperphosphorylation and aggregation, providing a foundation for therapeutic development. Given the significant unmet need for effective therapies for neurodegeneration, this study provides a foundation for developing restorative strategies targeting tau pathology.

RESULTS

Successful Cloning and Validation of Transgene

The correct cloning of the tau transgene into the AAV vector was confirmed using Agarose Gel Electrophoresis following SmaI digestion. The gel (*Figure 1*) exhibited distinct bands corresponding to the expected sizes (Supplementary Table 5), confirming the integrity of the inverted terminal repeats (ITRs) post-miniprep for Samples 1 and 3. Specifically, the clear and discrete bands observed in Samples 1 and 3 align with the predicted fragment sizes (two bands around 4000 bases, one band around 1000 bases and one band around 2000 bases) listed in Supplementary Table 5, indicating successful cloning and maintenance of essential vector structures.

All three samples were further validated by Sanger Sequencing with the primers hSyn and WRPE, which confirmed the sequence fidelity of the inserted tau transgene against the reference sequence. These results (*Figure 2*) collectively verified the successful cloning of the transgene into the vector. Each primer facilitated sequencing of approximately 800 base pairs, collectively providing coverage for around 1600 base pairs of the total 2307 base pairs of the gBlock. This sequencing strategy covered most of the transgene but left a central segment of approximately 707 base pairs unsequenced, which is highlighted in red in the Figure. The sequenced regions, shown in gray in the figure, matched the expected sequences of our assembled plasmid and the Twist gene fragment. However, thin red/orange lines in the figure indicate base mismatches within the sequenced portions.

Whole Plasmid Sequencing

Whole plasmid sequencing was conducted to verify the integrity of the entire vector sequence after cloning and midiprep processes. The sequencing data confirmed that the plasmid retained its structural integrity with no unintended mutations, affirming the accuracy of the cloning process and establishing the plasmid's suitability for subsequent experimental applications. The sequence alignments shown in *Figure 3A and C* demonstrate close agreement between the actual sequences obtained from samples and the constructed plasmid template, further validating the cloning accuracy. Interactive maps of the plasmid (*Figure 3B and D*) visually illustrate the complete structure, annotated with key features including the MAPT coding sequence, Ampicillin resistance gene (AmpR), origin of replication (ori), AAV2 ITRs, hSyn promoter, and the WPRE element. Each feature's

sequence alignment is displayed with percentage matches, where essential regions typically show high alignment accuracy between 98% and 100%, confirming that these critical sequences closely match the expected configurations. However, an anomaly in one of the AAV2 ITRs shows a match of only 19% due to fragmentation.

Additional sequence results (Supplementary Link 1) shows high quality sequencing. The .ab1 files exhibit clear peaks above 800 intensity, ensuring accurate base calling across essential genomic features like promoters and ITRs. Coverage data show depths between 400 and 600 reads per base, providing reliable and comprehensive sequencing coverage. Histograms indicate that most reads are full-length, reflecting the plasmid's integrity. Further analyses show 99.8% and 99.9% 1-mer formations, respectively, with minimal 2-mer formations and 1.0% E. coli contamination, which does not significantly affect plasmid utility.

Quantitative PCR of AAV Titer

The quantification of the AAV vector titre was assessed through quantitative PCR, with the results providing detailed insights into the viral concentrations and efficiencies. The amplification plot (*Figure 4*) demonstrated a typical exponential increase in fluorescence across multiple sample replications and dilutions, indicating successful PCR amplification of the targeted tau sequence within the AAV vectors. The melt curve analysis (*Figure 5*) further validated the specificity of the PCR products, displaying a distinct peak for the tau samples at the expected melting temperature (86-87°C). Additionally, a standard curve was generated (*Figure 6*), based on a series of known standard concentrations, showing a strong linear relationship ($R^2 = 0.9108$) between the cycle threshold (Ct) values and the logarithm of the initial concentrations. This curve was instrumental in calculating the exact viral titres in our samples. The Ct values obtained from the dilution series varied considerably, ranging from 19.598 to 24.246, indicative of logarithmic reductions in template concentration.

From the standard curve and Ct values, viral particle concentrations were computed using *Equation 1* and *2*. However, due to considerable variability in viral concentrations (*Table 1*) across the dilution series—from about 1,588,566.47 particles/µL at the lowest dilution to 20,074,757.69 particles/µL at the highest dilution—the decision was made to adopt the concentration at the lowest dilution factor as the physical titer for consistency in experimental applications. Consequently, the physical titer of our samples was determined to be 1.5 million

viral genomes (vg)/μL. This approach ensured a standardized concentration that was utilized to calculate the volumes of viral titre needed for the intended multiplicity of infections (MOI of 100 and 250) using *Equation 3* for precise and effective dosing in subsequent functional assays such as Western blots and staining procedures.

Functional Assay

Western Blot

The Western blot analysis intended to detect tau protein expression yielded inconclusive results. The image (*Figure 7*) exhibited blotchiness with no distinct bands visible for either the samples or the controls. Although the molecular weight ladder is somewhat discernible (with bands at about 50 kD), suggesting that the gel electrophoresis component functioned to some extent, the overall quality and clarity of the protein bands were compromised. This outcome may indicate issues with the protein transfer, antibody specificity, or sample quality.

Fluorescence Imaging

To investigate tau protein aggregation and the localization of FLAG-tagged tau in SKN disease cells, a dual-staining protocol was employed, combining Thioflavin staining for tau aggregates and anti-FLAG immunofluorescence to detect FLAG-tagged tau expression. Negative and positive control images confirmed the specificity of staining for both Thioflavin and FLAG-tagged tau, as shown in *Figure 8*. Cells were fixed with Neutral Buffer Formalin (NBF), permeabilized with Triton X-100, and blocked with normal goat serum and BSA to minimize nonspecific antibody binding. FLAG-tagged tau expression was visualized using a monoclonal anti-FLAG M2 antibody followed by Alexa Fluor® 594-conjugated goat anti-mouse IgG secondary antibody, which produced a red fluorescent signal. The FLAG signal localized to specific cellular regions, some of which overlapped with the Thioflavin-stained tau aggregates, as observed in the single-channel fluorescence images (*Figure 9* for MOI 100 and *Figure 10* for MOI 250). This co-localization suggests somewhat successful integration of FLAG-tagged tau into aggregated structures, although the overlap was not optimal, and visualization of the co-localization proved challenging; this will be further addressed in the Discussion section. DAPI staining produced blue fluorescence, providing identification of cell nuclei and enabling spatial mapping of tau protein localization and aggregate distribution. Overlaying the fluorescence channels revealed distinct

patterns of FLAG-tagged tau and tau aggregates, with regions of co-localization more pronounced at higher MOI levels as observed in *Figure 10*. In the merged fluorescence images, FLAG-tagged tau expression and tau aggregation were evaluated at two multiplicities of infection (MOI 100 and MOI 250). At MOI 100 (*Figure 11, Panel A*), FLAG-tau localization (TRITC, red) was observed in discrete regions, primarily adjacent to cell nuclei (DAPI, blue), with limited Thioflavin signal expected for early tau transduction. In contrast, at MOI 250 (*Figure 11, Panel B*), Thioflavin staining (FITC, green) revealed more widespread and intense tau aggregates alongside increased FLAG signal (TRITC, red), suggesting slightly enhanced tau transduction and aggregation with elevated viral genome delivery. The observed patterns support the relationship between MOI and tau pathology development, as higher MOI resulted in slightly greater tau protein expression and aggregate formation.

DISCUSSION

The successful cloning and validation of the tau transgene into the AAV vector were confirmed through multiple levels of analysis, ensuring the integrity of the vector and inserted transgene. Restriction digestion with *SmaI* and subsequent agarose gel electrophoresis yielded the expected band patterns, indicating correct integration and preservation of key vector elements such as the inverted terminal repeats (ITRs). The consistency of these results across Samples 1 and 3 suggests reliable cloning and efficient maintenance of vector structures during the miniprep process. Sanger sequencing provided additional validation, confirming sequence fidelity for most of the tau transgene. While the sequenced regions matched the expected sequence, a central segment of approximately 707 base pairs remained unsequenced due to primer limitations. This gap, while small, represents a minor limitation in the complete verification of the transgene sequence. Nevertheless, the absence of significant mismatches in the sequenced regions supports the accuracy of the cloning process.

Whole plasmid sequencing offered comprehensive validation of the vector's structural integrity post-cloning and midiprep. The sequence alignments revealed a high degree of agreement with the expected plasmid template, confirming that critical elements such as the tau coding sequence, hSyn promoter, WPRE, and ITRs were correctly assembled and maintained. Notably, an anomaly in one of the AAV2 ITRs, showing only a 19% match, suggests fragmentation or partial degradation of this element. However, this fragmentation is inherent to

the original plasmid backbone and thus does not impact the structural integrity of the plasmid. Thus, the overall sequence integrity was upheld, and the plasmid retained its suitability for subsequent experimental applications.

Quantitative PCR (qPCR) was employed to determine the viral titer of the AAV vector preparations. The amplification curves and melt curve analysis confirmed the specificity and efficiency of the qPCR assay, with a strong linear relationship observed in the standard curve. The single peak in the Melt Curve at 86-87°C indicates that there aren't any primer dimers, which can increase background signal and alter the Ct values of samples. However, considerable variability in Ct values across the dilution series was noted, leading to a range of viral particle concentrations. To address this variability, the decision was made to standardize the physical titer based on the lowest dilution factor, resulting in a final concentration of 1.5 million viral genomes (µg)/µL. The western blot analysis intended to detect tau protein expression yielded inconclusive results, with blotchiness and a lack of distinct bands for both the samples and controls. The presence of a discernible molecular weight ladder suggests that the electrophoresis step was partially successful; however, issues with protein transfer, antibody specificity, or sample quality likely compromised the overall outcome. It is important to note that prior to transduction, a significant issue was encountered in which the virus solution had aggregated. This aggregation is theorized to be due to residual purification matrix carried over from previous steps. To address this, Elution Buffer was used to break down the aggregates. While effective in dispersing the virus, Elution Buffer is not optimal for long-term storage in cells and may have impacted the stability and functionality of the viral particles. Thus, it is possible that the use of Elution Buffer to address virus aggregation affected the quality or quantity of protein expressed in the cells, hindering detection. Overall, further optimization is necessary to resolve these issues, and potential steps to do so include improving the protein transfer process, testing alternative antibodies, or optimizing the virus purification and storage protocols to maintain protein integrity.With Fluorescence Imaging, the lab group aimed to visualize tau protein aggregation and FLAG-tagged tau expression following AAV transduction at two multiplicities of infection (MOI 100 and MOI 250). The results demonstrated some degree of success in detecting tau aggregates and FLAG localization; however, several technical challenges impacted the quality and interpretation of the imaging data. One notable strength of this experiment was the successful detection of FLAG-tagged tau expression and tau aggregates

using Thioflavin and anti-FLAG staining. The TRITC (red) signal for FLAG-tagged tau, particularly at MOI 250, co-localized with FITC (green) Thioflavin staining in several regions, suggesting the presence of tau aggregates consistent with prior observations of tau pathology. The circular, punctate structures observed in the TRITC channel resemble granules, a hallmark feature of tau protein aggregation previously described in tauopathy studies[6]. This supports the conclusion that tau aggregation was achieved and visualized, albeit in a semi-quantitative manner.

However, several weaknesses in the imaging workflow limited the clarity and robustness of our results. A key technical issue was uncertainty regarding which side to mount the cells on the slide during the final preparation step. This ambiguity likely contributed to inconsistent focal planes, reducing the overall quality of the fluorescence images. Furthermore, while Thioflavin staining successfully highlighted tau aggregates, the DAPI signal used for nuclear visualization was notably weak. This was likely due to our use of ProLong Gold Antifade reagent containing DAPI instead of performing a dedicated Hoechst staining step for nucleic acids. Hoechst stains are known to provide stronger and more consistent nuclear signals compared to antifade DAPI formulations [7]. This shortcut ultimately compromised the clarity of nuclear visualization and the ability to spatially relate FLAG-tau expression and tau aggregates to the cell nuclei. Despite the limitations, FLAG-tagged tau was detected reliably across MOI conditions, and the Thioflavin staining confirmed the presence of aggregated tau structures. Notably, the tau signal at MOI 250 was more widespread and intense compared to MOI 100, consistent with higher viral genome delivery leading to greater tau expression and aggregation. However, the DAPI staining's poor performance hindered our ability to precisely determine cell morphology and accurately position aggregates relative to nuclei. Additionally, while the FLAG signal appeared robust, the granularity observed in the TRITC channel raises questions about the nature and distribution of tau aggregates. Tau staining typically reveals fibrillary or filamentous structures in advanced stages of tau pathology [7], whereas our results primarily showed circular puncta. This could indicate early-stage aggregation or technical artifacts in staining and imaging.

To address these challenges, several improvements can be implemented in future experiments. First, the mounting procedure for fixed cells should be standardized to ensure consistent focal planes and optimal imaging quality. Second, dedicated Hoechst staining should be performed for nuclear visualization, as this would provide stronger and more reliable nucleic acid staining compared to DAPI-containing antifade reagents. This would allow for better spatial analysis of tau aggregates relative to cell nuclei. In addition, further refinements to the imaging protocol are needed to verify the nature of the tau aggregates. Performing co-staining with additional tau-specific antibodies (e.g., AT8 or Tau-5) could help distinguish between early tau oligomers and more mature fibrillar aggregates. Advanced microscopy techniques, such as confocal imaging, could also improve resolution and provide clearer insights into tau structure and localization. Finally, increasing the sample size and exploring additional MOI conditions would help validate the relationship between viral genome delivery and tau aggregation more rigorously. While the results demonstrated FLAG-tau expression and evidence of tau aggregation, technical limitations such as weak nuclear staining, uncertain mounting orientation, and granular tau signals highlight areas for improvement. Despite these challenges, the experiment provides a foundation for understanding tau pathology and the role of FLAG-tagged tau in AAV-mediated transduction. Future studies incorporating optimized staining, imaging protocols, and additional molecular characterization will be critical for fully elucidating the behavior of tau aggregates in this system. For our functional assays as a whole, several technical challenges and methodological limitations may have influenced our experimental outcomes. A primary concern was the cell density variation between the Western blot and immunofluorescence plates, necessitated by the different plate formats. This deviation from the original protocol, where we intended to use 24-well plates for both analyses, was forced by the physical constraints of the coverslips. This inconsistency in cell numbers between plates (1×10^5 versus 2×10^4 cells/well) may have affected the viral transduction efficiency and subsequent protein expression levels. Additionally, the manual cell counting and seeding process could have introduced variability in cell numbers across wells, potentially contributing to inconsistent results. The AAV transduction protocol itself presented challenges, as the small volumes of viral solution required (as low as 1.3 μL) made accurate pipetting difficult and may have led to inconsistent MOIs across replicates. The 48-hour post-transduction timepoint, while standard, may not have been optimal for

detecting tau protein aggregation, as aggregate formation might require longer incubation periods. Furthermore, the Western blot analysis may have been complicated by potential protein degradation during sample preparation, despite the use of protease inhibitors. The immunofluorescence staining results could have been affected by high background fluorescence or non-specific binding of antibodies, making it challenging to distinguish genuine tau expression from artifacts. These technical limitations highlight the importance of protocol optimization and the need for additional controls in future experiments.

MATERIALS & METHODS

Cloning of Tau Transgene for AAV Expression

A Gibson Assembly cloning method was utilized to create the tau-expressing AAV. First, the tau protein sequence was obtained from UniProt [8] and codon-optimized [9] (see Supplementary Table 4 for sequences). The custom insert (gBlock) was then designed with several components: a Kozak sequence (GCCACC) as a ribosome binding site (RBS) to enhance translation initiation, an ATG start codon, and the codon-optimized tau transgene sequence. To enable easy detection and purification of the expressed tau protein, a FLAG epitope tag was added downstream of the tau coding region, with the sequence obtained from Cube Biotech [10]. The gBlock also included a TAA stop codon, concluding the transgene. At the same time, the pAAV-hSyn-mCherry-WPRE plasmid was chosen as the AAV backbone, which includes a human synapsin 1 (hSyn) promoter for neuron-specific targeting, a polyadenylation (PolyA) tail for RNA stability, a woodchuck hepatitis virus post-transcriptional regulatory element (WPRE) to enhance transgene expression, and an antibiotic resistance gene for selection. To introduce the tau gene, BamHI and EcoRI restriction sites within the backbone were utilized to excise the mCherry sequence, producing sticky ends to facilitate replacement with our codon-optimized tau sequence. Specifically, we mixed 25 µL of pAAV-hSyn-mCherry (120 ng/µL, totaling 3 µg), 3 µL of NEBuffer r3.1 (10x concentration), 1 µL of BamHI, and 1 µL of EcoRI to a final volume of 30 µL. This mixture was incubated at 37°C for 2 hours. Subsequent gel electrophoresis on a 0.8% agarose gel at 140 V for 40 minutes allowed us to isolate a 4.0 kb band, indicative of the correctly linearized vector, which was then purified using the Zymoclean Gel DNA Recovery kit. The purified linearized vector, now at a concentration of 129.9 ng/µL, was diluted to 50 ng/µL.

The linearized Tau DNA fragment (with unique overlaps), supplied by Twist, was resuspended to a concentration of 10 ng/uL and then ligated into the backbone using Gibson Assembly. First, 50 ng of the linearized plasmid backbone was used along with a 3-fold molar excess of the gBlock fragment insert, calculated using the NEBioCalculator [11]. After determining the concentrations of the plasmid and gBlock fragment (50 ng each), the appropriate volumes of each (1µL and 1.5 µL, respectively) were added to the reaction mixture. The reaction tubes contained 7 µL of Gibson assembly enzymes, buffers, and reagents; 3 µL of the DNA and water mixture was added to each tube to reach a final volume of 10 µL. For the experimental setup, a separate negative control was prepared, containing only the linearized plasmid backbone with no Twist fragment, replacing the insert volume with water. Both the experimental and control reactions were incubated at 50°C for 15 minutes in a thermocycler to facilitate the assembly.

***E. coli* Transformation with AAV Vector**

The assembled AAV vector was transformed into E. coli using the chemically competent heat shock protocol tailored for the STBL3 strain, which is ideal for vectors with repetitive sequences. First, two tubes of STBL3 cells were thawed on ice, and two LB+carbenicillin agar plates were warmed in a 37°C incubator, positioned upside down to prevent condensation. Next, 5 µL of the Gibson mix was added to each tube of thawed cells, and the contents were mixed by swirling the pipette tip, avoiding pipetting up and down to prevent damage to the fragile cells. The cells were then left on ice for 15 minutes before undergoing a 45-second heat shock at 42°C. This was immediately followed by a 2-minute stabilization period on ice to allow the cells to encapsulate the plasmid DNA. The contents of each tube were then evenly spread onto the warmed plates using a sterile L-spreader. The plates were incubated at 37°C for 16-20 hours, after which the colonies were inspected for growth (refer to Supplementary Figure 1) before the plates were transferred to a 4°C refrigerator to preserve the colonies for further analysis.

Sequencing Verification for Tau Transgene and AAV plasmid

Bacterial culture preparation and plasmid isolation via ZymoPURE Miniprep

3 colonies were inoculated into 5 mL of LB + carbenicillin media each by transferring with a sterile pipette tip, which was ejected into the culture tube. Cultures were incubated at 30°C for 12-16 hours in a shaking incubator. After which, archival glycerol stocks were created. Bacterial cultures exhibiting turbidity were processed for plasmid isolation using the ZymoPURE Miniprep kit. A 3 mL aliquot of bacterial culture was centrifuged at 15,000x g to pellet the cells, discarding the supernatant. Cells were resuspended in 250 µL of cold ZymoPURE™ P1 buffer by vortexing and lysed with 250 µL ZymoPURE™ P2 by gentle inversion. 250 µL of ZymoPURE™ P3 buffer was added to neutralize the solution, and it was centrifuged at 15,000 x g for 5 minutes to remove the lysate. 600 µL of the supernatant was mixed with 260 µL ZymoPURE™ Binding Buffer and loaded onto a Zymo-Spin™ II-PX Column attached to a vacuum manifold. Multiple washes were performed starting with 800 µL of Wash 1 and two 800 µL washes of Wash 2, the column was centrifuged to remove residual wash buffer, and the DNA was eluted with 25 µL of ultrapure water. DNA concentration and purity were determined using the NanoDrop spectrophotometer.

SmaI Digest and Agarose Gel Electrophoresis

Plasmid DNA (500 ng - 1 µg) was incubated with rCutSmart Buffer (final concentration 1x), SmaI enzyme (1 µL), and ddH2O was added to a final volume of 30 µL at 37°C for 15-30 minutes. A 1% agarose gel was prepared by combining 0.5 g agarose with 50 mL 1x TAE buffer, microwaving, and adding SYBR Safe DNA Gel Stain. After loading a DNA ladder and digested samples with 1x loading dye, gels were run at 120 V for 30 minutes and visualized with the iBright imaging system.

Sanger Sequencing and Data Analysis

Purified plasmid samples were prepared for Sanger sequencing using Genewiz universal primers according to Azenta Sample Submission Guidelines. Sequence analysis was conducted in Benchling using the .ab1 electropherogram files for alignment with the MAFFT algorithm, where the alignment quality and integrity of inserts were verified based on chromatogram peaks and sequence matching.

Bacterial culture preparation and plasmid isolation via ZymoPURE Midiprep

Bacterial cultures were prepared by inoculating a 5 mL LB medium (pre-supplemented with carbenicillin) with a small amount of glycerol stock of bacteria with sequence verified plasmids from -80°C storage. The frozen glycerol stock was scraped with a sterile pipette tip, and the tip was ejected into the LB medium and swirled to ensure transfer. The culture tube was loosely capped and incubated at 30°C for 5-8 hours in a shaking incubator. 5 mL of the starter culture was added to a 50 mL LB + Carbenicillin flask. This larger culture was returned to the incubator and shaken overnight for 12-16 hours to promote bacterial growth.

After confirming sufficient bacterial growth (based on the turbidity in the culture flask), plasmid DNA was isolated using the ZymoPURE™ Midiprep Kit. The bacterial culture was centrifuged to pellet cells and the pellet was resuspended in 8 mL of cold ZymoPURE™ P1 (Red) buffer by vortexing. 8 mL of ZymoPURE™ P2 (Blue) buffer was added, and the mixture was gently inverted multiple times to lyse cells (indicated by a purple, viscous solution). After a 2-3 minute rest at room temperature, 8 mL of ZymoPURE™ P3 (Yellow) buffer was added and inverted gently to neutralize the solution and produce a yellow precipitate. The lysate was loaded into a ZymoPURE™ Syringe Filter-X, and after a 5-8 minute wait, the precipitate was separated by pushing the lysate through the filter. Approximately 20 mL of cleared lysate was recovered and retained for the following binding step. 8 mL of ZymoPURE™ Binding Buffer was added to the clear lysate, mixed thoroughly by inversion, and applied to a Zymo-Spin™ V-PS Column Assembly on a vacuum manifold. Multiple washes were performed, the first one being with 5 mL of ZymoPURE™ Wash 1, followed by two washes with 5 mL of ZymoPURE™ Wash 2. After the final wash, the vacuum was kept on for an additional two minutes to remove residual liquid. The column was then centrifuged at 16,000 x g for 1 minute to remove any remaining wash buffer. The column was transferred to a clean 1.5 mL tube, and 200 µL of ultrapure water was added to the matrix center. After a 2-minute incubation, the sample was centrifuged at 16,000 x g for 1 minute to elute the plasmid DNA. Endotoxins were removed by loading the eluate into an EndoZero™ II Spin-Column and centrifuging at 10,000 x g for 1 minute, producing purified plasmid DNA. The concentration of the obtained DNA plasmid was measured using the NanoDrop.

Plasmidsaurus whole plasmid sequencing was used for verifying the quality of the transfer plasmid. The sample

for Plasmidsaurus sequencing was prepared by diluting the DNA concentration to the recommended 20-200 ng/μL range.

AAV triple transfection

AAV-293 cells (Agilent, Cat. No. 240073), a HEK293 cell line, was used for transfection. The cells cultured in high-glucose DMEM supplemented with 10% fetal bovine serum (FBS), 0.1mM MEM Non-Essential Amino Acids (NEAA), 2mM L-glutamine, and 1% Pen-Strep at 37°C with 5% CO_2. Cells were passaged 2-3 times before transfection to ensure optimal cell health and density. The transfection complex was prepared in 2 1.5 mL microcentrifuge tubes. The total DNA added to each microcentrifuge tube was 28 μg, maintaining a molar ratio of approximately 1:1:1 for each of the plasmids. 4.3 μg of Polyethyleneimine (PEI) was added per μg of DNA, summing up to 120 μL of 1 mg/mL PEI for each tube. After vortexing for 5-10 seconds, the solution was incubated at room temperature for 5-10 minutes to allow complex formation. Two 15 cm culture dishes with 70-80% confluence were prepared for transfection. Each plate contained 14.5 μg pXX680 helper plasmid (adenoviral helper genes), 11.2 μg AAV vector plasmid (AAV transfer vector) and 8.7 μg AAV helper plasmid (AAV RepCap vector) and 0.5 mL. The plates were prepared by adding the transfection complex solution dropwise and swirling the medium to ensure even mixing. The culture plates were incubated at 37°C and 5% CO_2 for 48 hours post-transfection. Cells were then harvested.

AAV lysis and harvest

After 48 hours, the cells were lysed using 10X AAV-Max lysis buffer (ThermoFisher Scientific, Cat. No. A50520) supplemented with Benzonase (MilliporeSigma, Cat. No. 1.03773.1010) and incubated at 37°C in the CO_2 incubator for 2 hours. Cells were scraped and transferred to a centrifuge tube and centrifuged for 30 minutes at 4500 rpm at 4°C. The supernatant and pellet were collected and stored at -80°C.

AAV Titering using qPCR

To determine the physical titer of AAV samples, a serial dilution of the pAAV-CAG-GFP plasmid standard curve (2×10^9 molecules/μL) was performed. Purified AAV samples were treated with DNase I (5 μL rAAV, 39 μL H_2O, 5 μL 10X DNase buffer, 1 μL DNase) and incubated at 37 °C for 30 minutes. The DNase-treated samples were serially diluted, and technical duplicates of four dilutions were analyzed. A qPCR master mix

containing SYBR Green, Forward and Reverse ITR primers, and nuclease-free water was prepared, and 15 µL of this mix was added to 5 µL of sample/standard in each well. The qPCR plate was run on a QuantStudio3 instrument. Data analysis including standard curve validation (R^2 ~1.0, PCR efficiency 90–110%), melt curve analysis (single peak), and assessment of Ct value consistency among duplicates (≤0.5 Ct difference) was performed. The results were used to calculate viral genomes (vg)/mL based on the standard curve.

Cell Seeding and Transduction

For protein expression analysis via Western blot, SKN cells were seeded in a 24-well plate at a density of 1×10^5 cells per well in 500 µL of complete DMEM medium (supplemented with 10% FBS, 1% Pen-Strep). For immunofluorescence staining, cells were seeded in a 12-well plate containing sterile glass coverslips at a density of 2×10^4 cells per well in 1 mL of complete DMEM medium. Both plates were incubated at 37°C with 5% CO_2 for 24 hours to achieve approximately 70-80% confluence before transduction. The experimental design included two technical replicates for each condition: positive control (untransduced), negative control (empty vector), and two different multiplicities of infection (MOI) - 100 and 250 (see Figure 12 for plate layout). The required volume of AAV for each MOI was calculated based on the viral titer and cell number per well. For the 24-well plate (Western blot), 6.5 µL and 16.25 µL of AAV were used for MOI 100 and MOI 250, respectively. For the 12-well plate (immunofluorescence), 1.3 µL and 3.25 µL of AAV were used for MOI 100 and MOI 250, respectively, accounting for the different cell numbers between plates. These volumes for the selected MOIs were calculated referring to Equation 3. Transduction was performed by directly adding the calculated volumes of AAV to each well containing fresh complete DMEM medium. The plates were gently swirled to ensure even distribution of the viral particles and returned to the incubator. Cells were maintained at 37°C with 5% CO_2 for approximately 48 hours post-transduction before proceeding with protein extraction or immunofluorescence staining procedures.

Protein Extraction

Cells were washed with PBS twice and lysed with Radioimmunoprecipitation assay (RIPA) buffer with protease inhibitors (100:1 ratio). 75 µL of RIPA buffer was added per well of the 24 well-plate. Cells were incubated on

ice for 5 minutes and centrifuged at 16,000g for 10 minutes. Supernatant containing the extracted proteins was collected and the protein concentration was measured.

Protein Preparation and SDS-PAGE

Cell lysates were thawed on ice, and protein samples were prepared by mixing appropriate volumes of protein sample (20 µg in this case), water, and 4x Laemmli buffer containing β-mercaptoethanol (10µL) to achieve a final protein concentration of ~0.67 µg/µL in a 40 µL volume. Samples were denatured by heating at 95°C for 15 minutes, then briefly centrifuged. SDS-PAGE was performed using pre-cast gels and 1x Tris-Glycine SDS running buffer. A total of 30 µL of each denatured protein sample and 5 µL of protein ladder were loaded into designated wells. Gels were run at 180 V for 50 minutes until the dye front reached the bottom.

Western Blot

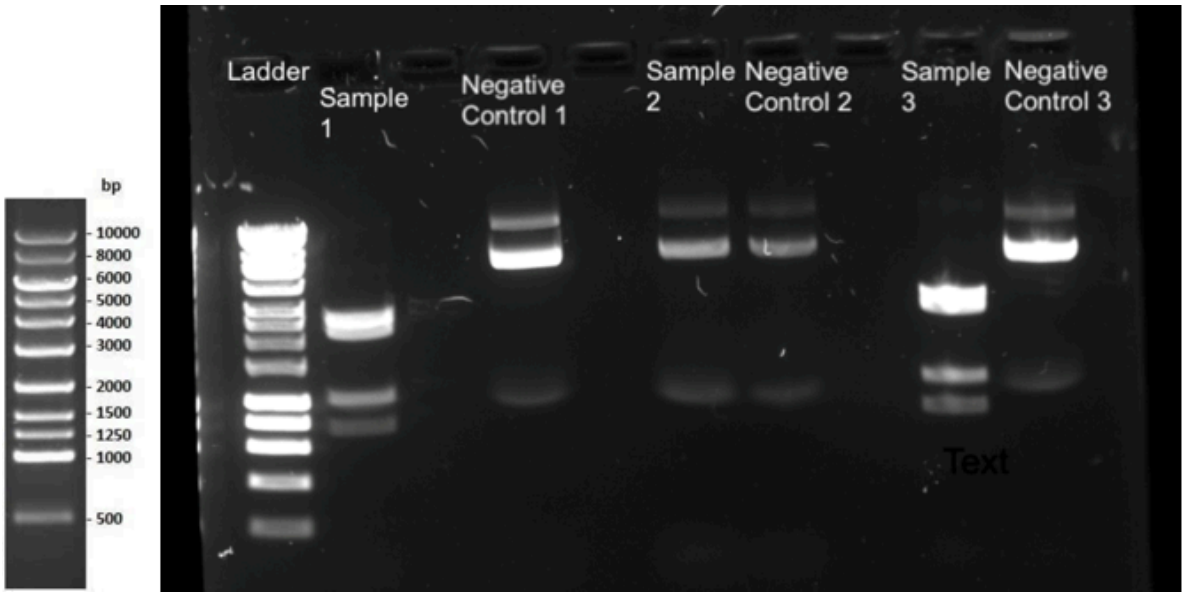
Protein samples were separated by SDS-PAGE and transferred to a PVDF membrane. Membranes were pre-activated with methanol, rinsed in distilled water, and incubated with 1X iBind™ Flex Solution for blocking. The membrane was loaded onto an iBind™ Flex Card, and the card was assembled into the iBind™ Flex Western Device. Diluted primary antibodies - Anti □-actin antibody (abcam: ab8227) and F1804 anti-FLAG M2, a monoclonal anti-mouse antibody (SigmaAldrich UNSPSC Code:12352203) in iBind Flex solution (1:1000) and secondary antibodies - Goat anti-Rabbit IgG (H+L) Secondary Antibody, HRP (Catalog # -31460 and Goat anti-Mouse IgG Alexa Fluor 594 (ThermoFisher Catalog # A-11032) were prepared in 1X iBind™ Flex Solution (1:4000), and were sequentially loaded into designated wells along with a washing buffer as outlined in the protocol [12]. The device was run for ~23 hours, allowing automated blocking, antibody binding, and washing via sequential lateral flow. Following the run, the membrane was rinsed and incubated with chemiluminescent substrate for signal development, and imaging was performed using a digital imager.

Staining

In the staining protocol adapted from Thermo Fisher's 3 Color Staining protocol for Alpha-Catenin in HeLa Human Cervical Cancer Cells [13], tau protein detection in fixed cells was achieved through a series of preparatory and staining steps. Initially, the media was removed from live cells which were then fixed for 15

minutes using Neutral Buffer Formalin (NBF), with each well receiving 300 µL of the fixative to ensure coverage. Subsequent to fixation, cells were washed three times for 5 minutes each in PBS (1X). Permeabilization was facilitated by treating cells for 10 minutes with PBT (0.1% Triton X-100 in PBS), followed by another series of three 5-minute washes in PBS. For cell staining, the cells were first blocked for 1 hour using an antibody blocking buffer composed of 10% normal goat serum (NGS) and 1% BSA in PBT. This was followed by a 45-minute incubation with the primary antibody, F1804 anti-FLAG M2 - monoclonal anti-mouse antibody, at a concentration of 5 µg/ml in antibody blocking buffer, specifically targeting the FLAG tag linked to the tau protein. After primary antibody incubation, cells were washed for 15 minutes in PBT. A secondary antibody, Alexa Fluor® 594 goat anti-mouse IgG, was then applied at 5 g/ml in 1% BSA/PBT for 30 minutes to facilitate fluorescence detection. Afterward, cells underwent three 5-minute washes in PBS. To detect tau protein aggregation, cells were further treated with Thioflavin, diluted 1:40 from the stock solution, and incubated for 20 minutes. This was followed by three additional 5-minute washes in PBS. For mounting, ProLong® Gold antifade reagent, which contains DAPI stain aiding in nuclear visualization, was used. This reagent was equilibrated to room temperature before application to avoid any potential degradation from external heat sources. Specimens were then mounted by applying 1 drop of the antifade reagent directly onto the slide and another 2 drops on the coverslip with the fixed cells. Finally, the samples were allowed to cure on a flat surface in the dark.

FIGURES



[14]

Figure 1. Agarose Gel Electrophoresis of SmaI Digest of AAV Vector Constructs. This image shows the molecular weight ladder in the first lane followed by alternating samples and negative controls. The gel illustrates distinct bands at the expected lengths for Samples 1 and 3, indicating successful digestion and integrity of the transgene within the vectors. Sample 2 and the negative controls do not show expected digested distinct bands, highlighting issues with either the digestion or the sample quality.

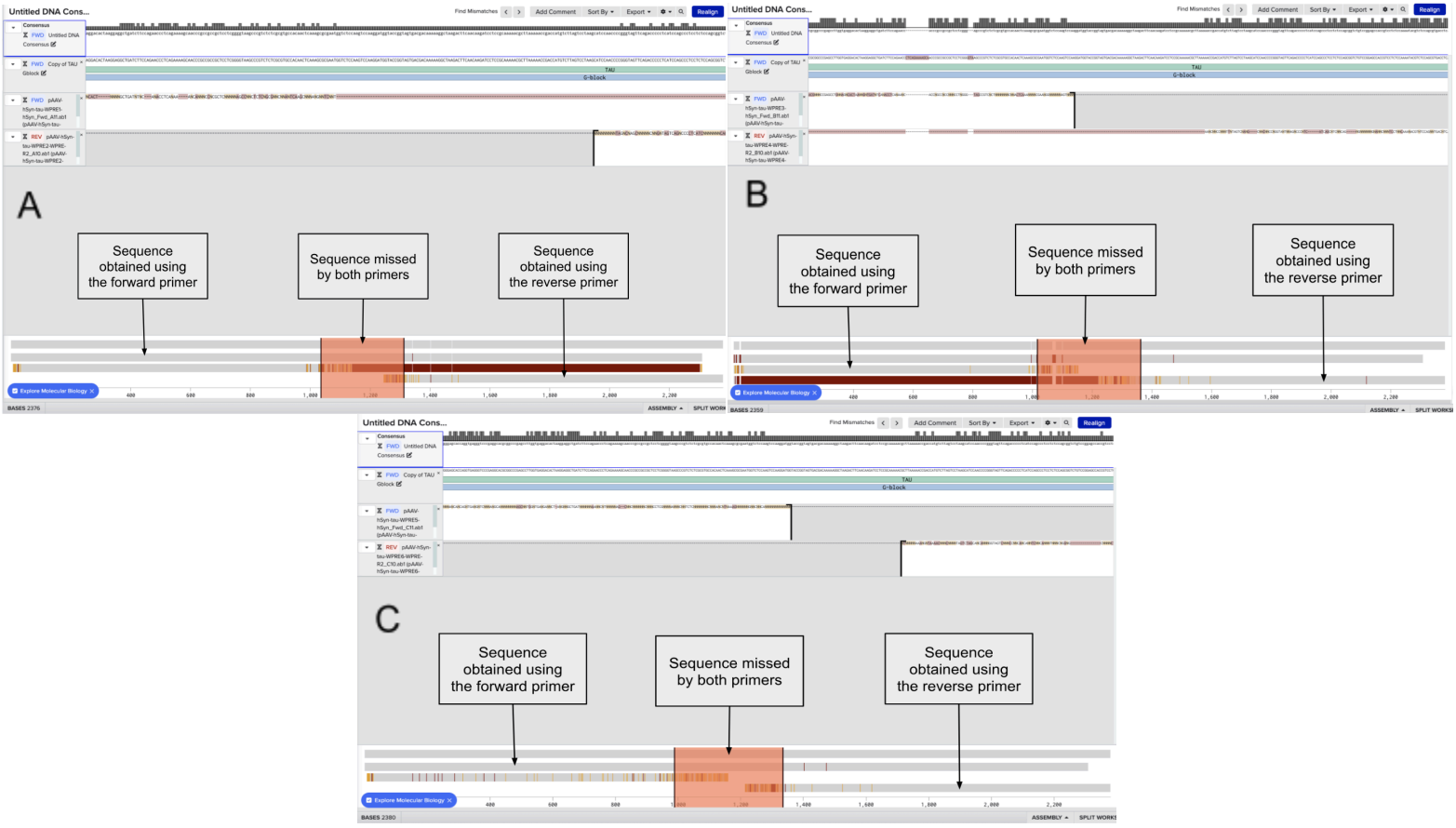


Figure 2: Sequencing Alignment Results of the AAV Vector Constructs. The alignment showcases sequences from the tau transgene insert regions, with gray regions indicating sequence matches to the reference, and thin red/orange lines representing base mismatches. The absence of sequence data in the middle portions (highlighted in red) is due to the unsequenced central segment of the transgene, with the available sequences affirming the overall fidelity of the cloned sequences. (A) Sample 1 (B) Sample 2 and (C) Sample 3.

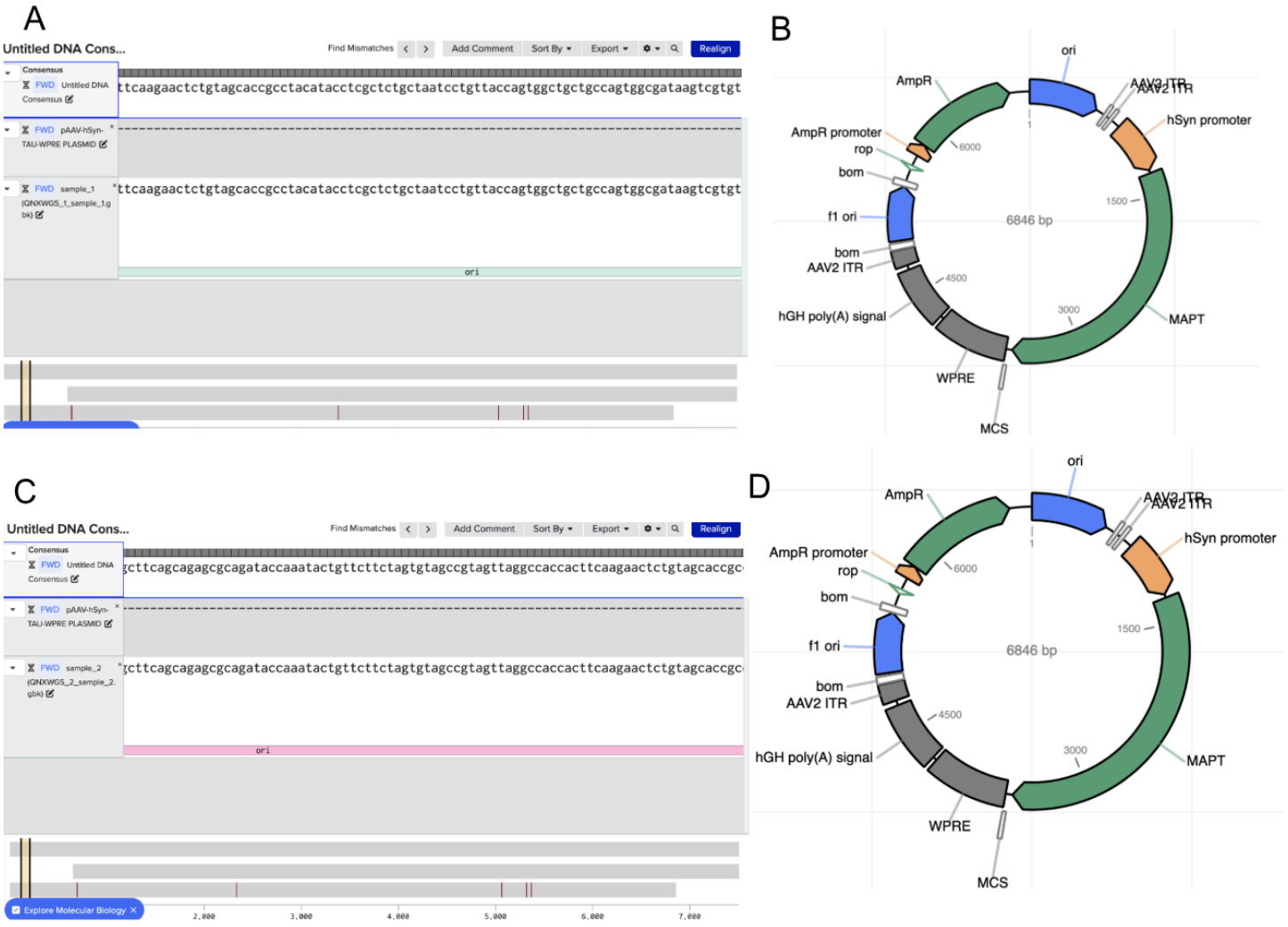


Figure 3: DNA Consensus and Alignment Analysis. Panel A and C show DNA sequence alignments between samples (1 and 2) and a constructed plasmid template, highlighting regions of high sequence fidelity and areas of discrepancy. Panels B and D depict interactive plasmid maps illustrating the structural annotations of the AAV vector, including key elements like the Ampicillin resistance gene (AmpR), human synapsin (hSyn) promoter, AAV2 inverted terminal repeats (ITRs), and the woodchuck hepatitis virus post-transcriptional regulatory element (WPRE). These maps provide a visual overview of the plasmid's organization and are essential for verifying the correct assembly of genetic components.

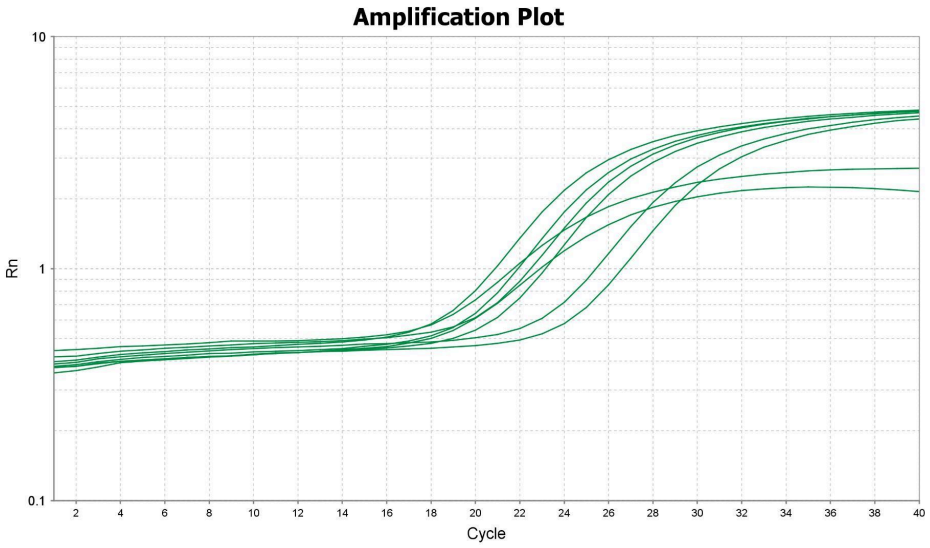


Figure 4: Amplification Plot for Quantitative PCR of AAV Titres. Amplification plot displaying the exponential increase in fluorescence across multiple cycles, indicating successful amplification of tau sequences.

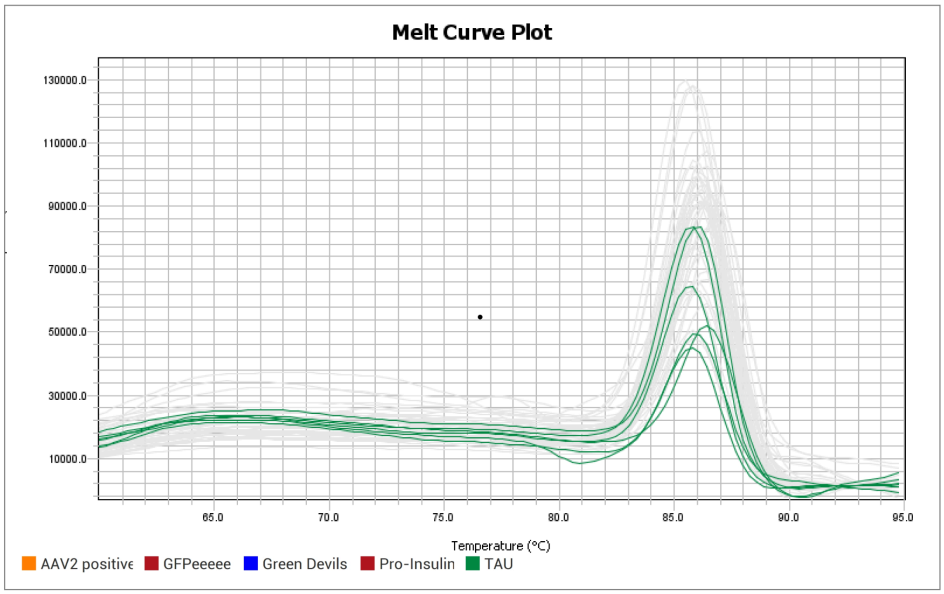


Figure 5: Melt Curve Analysis for AAV Vector Quantitative PCR. Melt curve analysis verifying the specificity of the PCR products. The curve shows a distinct peak at the expected melting temperature (86-87°C) for the tau sequences, confirming the absence of nonspecific products or primer-dimers.

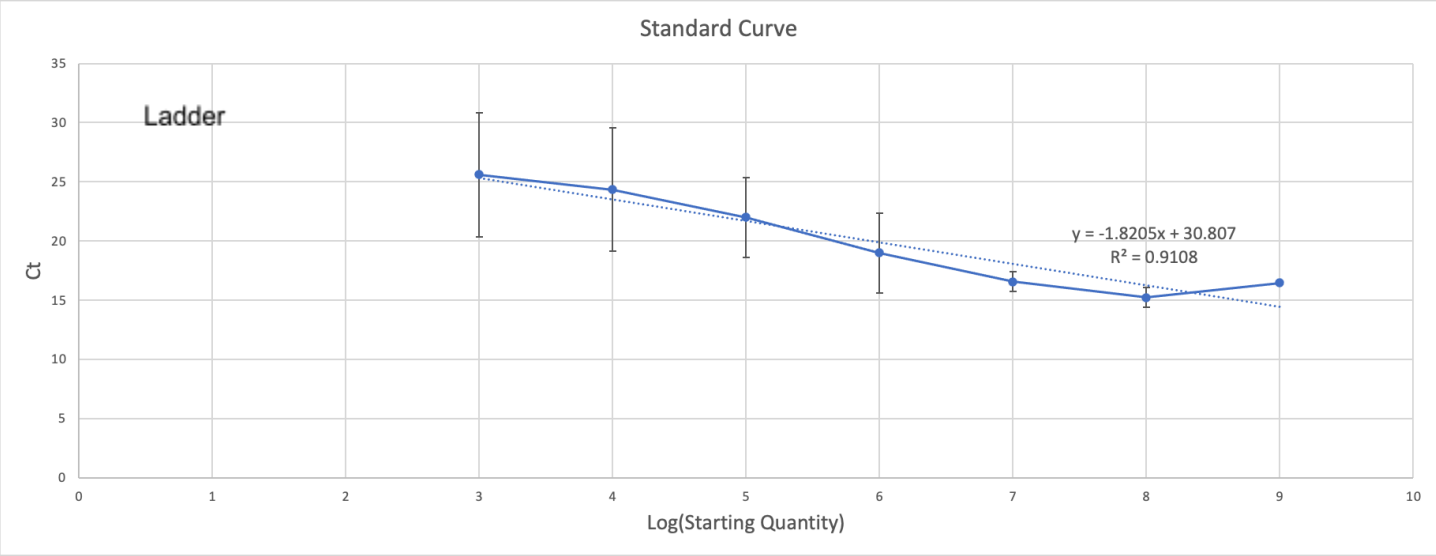


Figure 6: Standard Curve for Quantitative PCR of AAV Titres. Standard curve demonstrating the linear relationship between the logarithm of known standard concentrations and the observed cycle threshold (Ct) values. The curve, with an R^2 value of 0.9108, was used for determining the viral concentrations in sample dilutions.

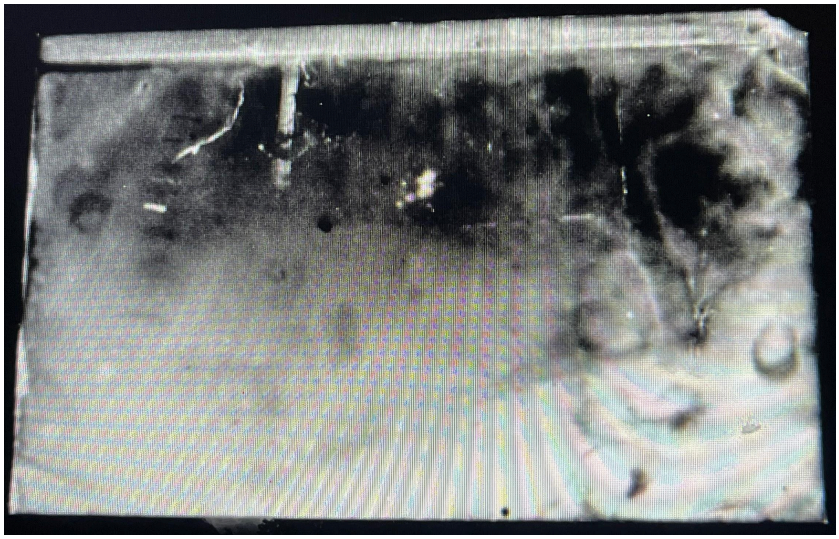


Figure 7: Western Blot Analysis for Tau Protein Detection. The blot shows significant blotchiness and lacks distinct protein bands for both samples and controls, suggesting issues with protein transfer or sample quality. While the molecular weight ladder is faintly visible, the absence of clear bands limits the interpretation of tau protein expression.

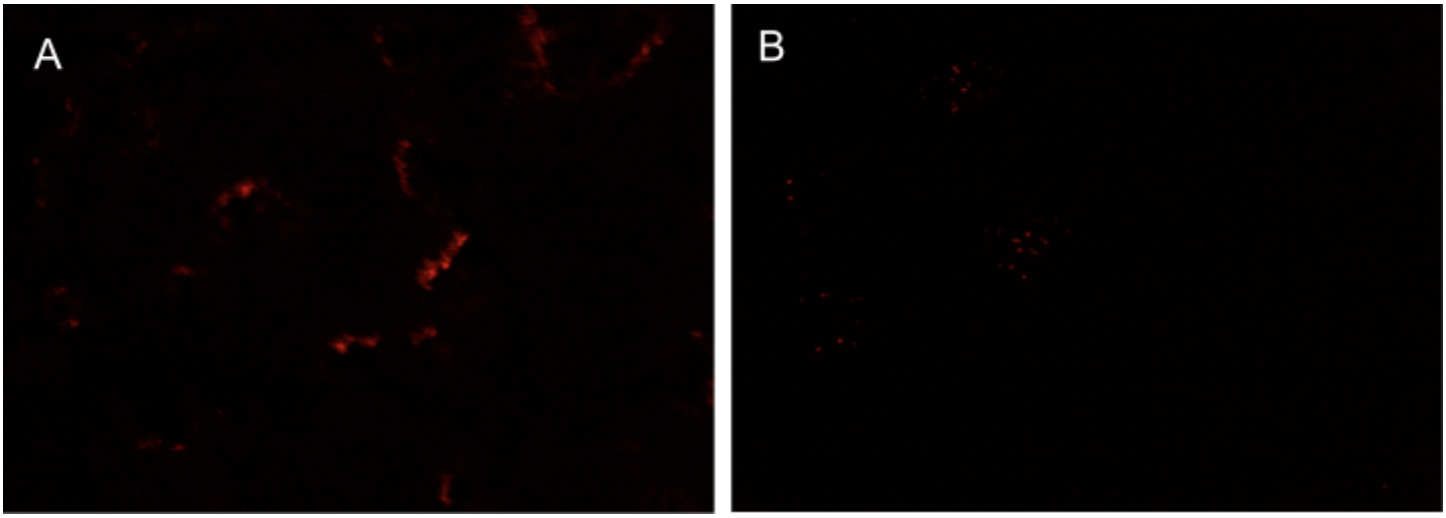


Figure 8. Validation of FLAG-tagged tau staining using TRITC fluorescence in negative and positive controls. (A) Positive control: Specific FLAG staining (red) is observed in discrete regions, confirming the presence of FLAG-tagged tau expression. (B) Negative control: Minimal background signal with no visible FLAG staining, demonstrating the specificity of the anti-FLAG antibody and TRITC secondary detection. These controls validate the accuracy and reliability of the staining protocol used for subsequent experimental analyses.

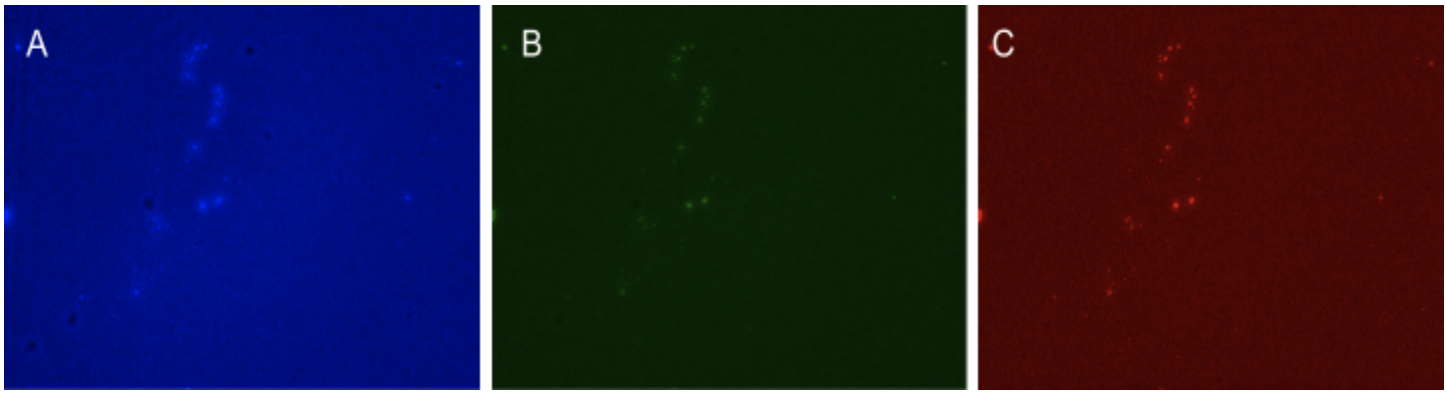


Figure 9. Single-channel fluorescence imaging of SKN disease cells transduced at MOI 100. (A) DAPI staining (blue) highlights cell nuclei, providing spatial reference. (B) Thioflavin staining (green) shows tau aggregates, visualized as discrete fluorescent puncta. (C) Anti-FLAG immunostaining (red) reveals FLAG-tagged tau localization. The presence of both Thioflavin and FLAG signals suggests FLAG-tagged tau expression with partial co-localization in aggregated regions.

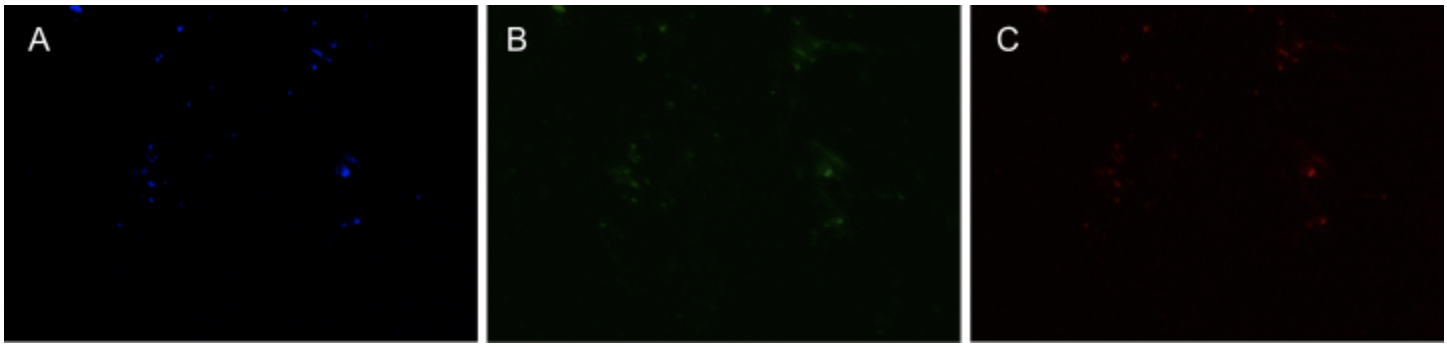


Figure 10. Single-channel fluorescence imaging of SKN disease cells transduced at MOI 250. (A) DAPI staining (blue) marks cell nuclei, providing a structural reference. (B) Thioflavin staining (green) highlights tau aggregates, appearing as more pronounced and widespread punctate structures compared to MOI 100. (C) Anti-FLAG immunostaining (red) reveals FLAG-tagged tau localization, with increased signal intensity and distribution. The enhanced Thioflavin and FLAG signals at higher MOI suggest increased tau expression and aggregate formation.

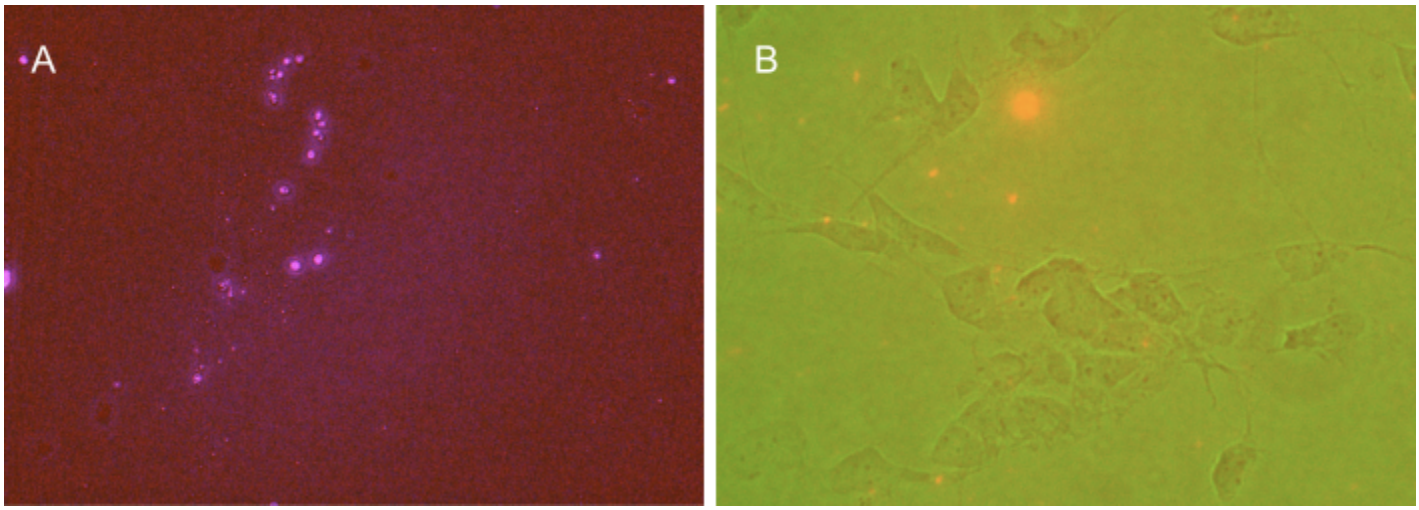


Figure 11. Merged fluorescence channels showing tau pathology and FLAG-tagged tau expression at MOI 100 and MOI 250. (A) MOI 100: DAPI (blue) and TRITC (red) channels are merged to highlight cell nuclei and FLAG-tagged tau localization, with FLAG signal observed in discrete regions alongside nuclei. (B) MOI 250: FITC (green) and TRITC (red) channels are merged to show Thioflavin-stained tau aggregates and FLAG-tagged tau, respectively. The higher MOI results in increased Thioflavin signal intensity and more widespread FLAG-tau distribution, suggesting enhanced tau aggregation and expression at elevated viral genome delivery.

Western Blot

1e5 cells/well

Positive Control	Negative Control	MOI 100	MOI 250		
Positive Control	Negative Control	MOI 100	MOI 250		
Positive Control	Negative Control	MOI 100	MOI 250		

Staining

2e4 cells/well

Positive Control	Negative Control	MOI 100	MOI 250
Positive Control	Negative Control	MOI 100	MOI 250
Positive Control	Negative Control	MOI 100	MOI 250

Figure 12. Experimental layout for Western blot and staining procedures at varying multiplicities of infection (MOI). The table summarizes experimental conditions used for Western blot analysis with 1e5 cells/well and staining with 2e4 cells/well. Both experiments included positive controls, negative controls, and two MOI conditions (MOI 100 and MOI 250) to evaluate FLAG-tagged tau expression and tau aggregation. Experimental duplicates were performed to ensure reproducibility across conditions.

TABLES

Table 1: Quantitative PCR Results and Calculated Viral Concentrations Summary of cycle threshold (Ct) values from a dilution series, used to calculate the number of viral particles and the concentration of viral particles (particles/µL).

Dilution Factor	Average Ct Value	Number of Viral Genomes (vg)	Concentration of Viral Genomes (vg/uL)
10	20.0255	7943282.347	1588656.47
200	19.626	277171966.8	55434393.4
1000	21.065	224531419.3	44906283.9
5000	24.2465	20074757.69	4014951.54

EQUATIONS

Equation 1: **Quantitative PCR standard curve equation used to calculate the logarithm of viral copies.** The parameters 'a' and 'b' are derived from the linear regression of the standard curve, where 'a' is the y-intercept and 'b' is the slope.

$$Ct = a + b(log\ copies)$$

Where, $a = 30.807$ and $b = -1.8205$

Equation 2: **Formula for calculating the concentration of viral particles (vg/µL).** This equation transforms the logarithmic copy number obtained from Equation 1 into an absolute number of viral genomes per microliter, adjusted for the dilution factor used in the sample preparation.

$$Viral\ Particle\ Concentration\ (vg/uL) = 10^{log\ copies} \times Dilution\ Factor$$

Equation 3: Formula for calculating the volume of AAV titre to add to cell cultures. This equation determines the appropriate volume of AAV vector needed to achieve a specified multiplicity of infection (MOI), based on the number of cells in the well and the known concentration of viral particles (vg/μL).

$$Volume\ of\ AAV\ titre\ to\ add\ to\ cells = \frac{(MOI) \times (number\ of\ cells\ in\ well)}{Viral\ Particle\ Concentration}$$

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AUTHOR CONTRIBUTIONS

All members performed laboratory experiments together. All members selected the gene therapy construct of interest, performed the molecular assembly and *E.coli* transformation, and executed the plasmid DNA miniprep, ITR validation, and subsequent Sanger Sequencing. I.M. carried out the midiprep, but whole plasmid sequencing was done with all members. All members conducted the AAV triple transfection and harvest, purification, and titering using quantitative PCR. For our own experimental design, I.M, S.M. prepared the plate layout for both functional assays, I.M. seeded the cells onto the plates, all group members performed the transduction, and N.K., K.W. completed the Western Blot while I.M., S.M. undertook cell staining and mounting for imaging purposes. I.M. imaged the cells at all necessary fluorescent channels. All members contributed equally to the manuscript, and each one thoroughly proofread the entire document.

DECLARATIONS

The authors have nothing to declare.

COMPETING INTERESTS STATEMENT

The authors have no competing interests.

DATA AVAILABILITY & CODE AVAILABILITY

Supplementary Information: Link

Supplementary Tables: Link

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